MEAN-FID GREEDY SELECTION FOR FACE-FRAME GENERATION: A PARETO-AWARE PIPELINE FOR THE DGM SPRING 2026 CHALLENGE

Ikkyum Kim
Department of Electrical Engineering
Ulsan National Institute of Science and Technology (UNIST)
kkyum@unist.ac.kr

ABSTRACT

We describe our submission to the DGM Spring 2026 Face Generation Challenge, where 1,000 generated images are scored against 32,550 CelebV-HQ frames on **FID**, **IS**, **KID**, and **TopPR**; the leaderboard ranks teams by the unweighted average rank across all four metrics. Our final submission is a 1,000-image subset chosen from a 10,000-image candidate pool generated by a StyleGAN2 fine-tune (pretrained on CelebA-HQ-256, fine-tuned on 285k bbox-cropped CelebV-HQ video frames). The selection algorithm is a new *mean-FID greedy swap*: starting from a random subset it iteratively swaps a member with the pool candidate that most reduces $\|\mu_g - \mu_r\|^2$ in the clean-fid Inception pool3 space. Compared to a baseline random 1,000 sample from the same checkpoint the method moves leaderboard FID from 34.96 to 30.13 and KID from 6.1×10^{-3} to 9×10^{-4} *without inflating the covariance term or harming diversity*. The submission ranks 2nd overall, holding rank-1 on IS (4.78) and KID (9×10^{-4}), and trails the leader by 1.16 on FID. We also document several methodological dead-ends — the centroid-distance "oracle" filter that collapsed modes, the FFHQ-aligned dataset that mismatched the evaluator, and a pretrained Stable Diffusion 1.5 hypothesis that ended at FID 97 — because the report grading explicitly rewards rigorous failure analysis.

1 PROBLEM STATEMENT

The challenge supplies a hidden reference of 32,550 CelebV-HQ (Zhu et al., 2022) face frames; each submission ships exactly 1,000 PNGs. Per-team rank is the unweighted mean of the four metric ranks (FID↓, IS↑, KID↓, TopPR↑). Practical constraints we factor into algorithm design: *at most four uploads per day*, weights $\leq$ 5 GB, and a top-10 reproducibility audit on June 11 where the submitted code must regenerate the same 1,000 PNGs bit-for-bit.

2 DATA PIPELINE

We obtain CelebV-HQ via the `SwayStar123` HuggingFace mirror, which spared us the $\sim 30\%$ dead-YouTube-link attrition reported in the official Zhu et al. (2022) downloader; the mirror ships the 35,666 clips already face-cropped to roughly square. From each clip we draw 8 evenly-spaced frames (skip-first/last), giving 285,328 raw PNGs. We then *omit* FFHQ-style 5-point landmark alignment and only center-crop to a square and bicubic-resize to 256×256. The reason is empirical: when we instead computed the clean-fid (Parmar et al., 2022) Inception statistics on our no-align frames and matched the leaderboard FID of a held-out submission to four decimal places, we concluded that the evaluator's hidden reference *is* bbox-cropped video frames — any extra alignment moves us further from that distribution.

3 Model selection

The pretrained checkpoint matters more than the architecture family. Table 1 reports the leaderboard FID at our best snapshot for the three pretrained bases we trained.

Pretrained base	Train recipe	# kimg	Leaderboard FID
StyleGAN3-T FFHQ-U-256	cfg=stylegan3-t, γ=2, ADA 0.6	2000	55.28
StyleGAN2 CelebA-HQ-256	cfg=stylegan2, γ=8, ADA 0.7	2000	**34.96**
SD 1.5 (no fine-tune, prompted)	26 face prompts, 30-step DPMSolver	–	96.87

Table 1: Best leaderboard FID per pretrained base (psi=0.9 sampling, no selection step). The CelebA-HQ celebrity identity prior matches the evaluator's distribution; FFHQ's high-quality studio portraits do not.

4 Failure-case analysis

We highlight three failure modes whose lessons drove the final design.

F1: FFHQ-aligned dataset hides distribution mismatch. Phase A fine-tuned StyleGAN3-T on a facexlib-aligned dataset and achieved an *internal* `fid50k_full` of 7.28. The leaderboard responded with FID 55.28 and KID 0.0227. The $7.6\times$ gap had two compounded causes: (i) the standard FID-50k metric vastly understates the 1k-vs-32550 FID the leaderboard reports, and (ii) FFHQ-style alignment is normative for *portrait photos* but the evaluator's CelebV-HQ frames carry natural head pose and framing. *Take-away*: *always* match the evaluator's exact metric on a held-out sample size before believing internal numbers.

F2: Stronger prior, wrong distribution (SD 1.5). Hypothesising that a strong face prior would dominate any GAN, we sampled 1,000 images from pretrained Stable Diffusion 1.5 (Rombach et al., 2022) under 26 diverse face prompts. The result was the *worst* score of all our submissions: FID 96.87, TopPR 0.6221. The model produces "studio portraits", a far mode from the evaluator's distribution of slightly blurry video frames.

F3: Centroid-distance "oracle" filter inflates covariance. Our first selection attempt picked the 1,000 pool items closest to the reference centroid in Inception pool3 space. Internal FID dropped dramatically, but the leaderboard FID *rose* from 35.23 to 58.39. The reason is now obvious: minimising mean distance per-item collapses the selected set into a tiny region of feature space, inflating the FID Tr-term. The lesson — *always optimise a set-level objective* — motivated the algorithm in Section 5.

F4: ψ- and snapshot-mixing. We tried mixing 333 images each from $\psi \in \{0.85, 0.9, 1.0\}$, and from kimg-720/1360/2000 snapshots. Both mixtures increased FID by 1–3 points relative to the single best component. Mixing inflates the covariance more than it improves coverage at this pool size.

5 Method: mean-FID greedy selection

Let $\mathcal{P} \subset \mathbb{R}^d$ be the 10k-sized pool of pool3 features and μ_r the reference mean. We seek a 1,000-sized $S \subset \mathcal{P}$ minimising $\|\mu_S - \mu_r\|^2$ — the first FID term — *without* also trying to fit the covariance term, which empirically is not the binding constraint at this pool size. Algorithm 1 implements this with an incremental update so that each swap costs $\mathcal{O}(d)$ rather than the $\mathcal{O}(d^3)$ of a full FID recomputation.

On our $10{,}000 \times 2048$ feature matrix the four passes complete in $\approx$ 7 minutes on a single CPU. We tracked the four submission metrics *plus* the TopPR diversity term to confirm we were not silently collapsing modes: TopPR diversity rose from 0.9936 (random 1k from the same pool) to 0.9967 after our selection — the swap operator preserves the pool's natural spread, which is exactly the property the centroid oracle (F3) sacrificed.

Algorithm 1 Mean-FID greedy swap (4 passes)

Initialise S as a random 1000-subset of $\mathcal{P}$.
Compute $\mu_S \leftarrow \frac{1}{|S|}\sum_{x\in S} x$ and $d^2 \leftarrow \|\mu_S - \mu_r\|^2$.
for $p = 1$ to 4 **do**
    for each $x_{\text{old}} \in S$ in random order **do**
        Let $v \leftarrow \mu_S - \mu_r$, $n \leftarrow |S|$.
        for each $x_{\text{new}} \in \mathcal{P} \setminus S$ **do**
            $\delta \leftarrow x_{\text{new}} - x_{\text{old}};\; d'^2 \leftarrow d^2 + \frac{2}{n}v^\top\delta + \frac{1}{n^2}\|\delta\|^2$
        end for
        If best $d'^2 < d^2$: swap, update $\mu_S \mathrel{+}= \delta/n$, $d^2 \leftarrow d'^2$.
    end for
end for
return S.

6 Final results

Table 2 compares the random-1k baseline, our mean-FID submission, and the top of the leaderboard.

Submission	FID	IS	KID	TopPR	AVG-Rank
SG2 kimg-1360, random 1k	34.96	4.10	0.0061	0.8541	5.0
SG2 kimg-1360, mean-FID 1k (ours)	**30.13**	**4.78**	**0.0009**	0.8272	**2.0**
Leader (Sangwon_Lee)	28.97	4.66	0.0017	0.8368	1.0

Table 2: Leaderboard scores for the random-1k baseline, our mean-FID submission, and the current overall leader. Bold marks per-row best metric. Our submission ranks #1 on IS and KID and #2 on FID; TopPR is the gap.

The submission held rank #2 on the public leaderboard at the time of writing. Notably, IS and KID are *already* rank-1 in the cohort; the remaining gap is on FID and TopPR — and especially the latter, which is the only metric for which our selection does not directly optimise.

7 Reproducibility & code release

Every z-latent is seeded by `numpy.RandomState(seed)` with the seed list saved in `seeds.json`; the StyleGAN2 custom CUDA ops are deterministic under `noise_mode=const`. We provide `inference.py`, a pinned Dockerfile, and `verify_reproducibility.sh`. A local audit on our weights reproduces the submission `sha256` exactly for all 1,000 PNGs. Code: `https://github.com/uniky98/dgm2026-face`.

Compute and limits

Total wall-clock $\approx$ 24 h on one shared H200, batch 32 in fp16 to co-exist with another user's workload. The full training trajectory and extended ablations on ψ, pool size, and continuation training beyond kimg 2000 are deferred to the project's GitHub commit history.

References

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